THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** human red blood cell	more space for haemoglobin
2	**(c)** nucleus	contains DNA
3	**(d)** totipotency	Haberlandt
4	**(c)** tendons	strong cord-like
5	**(b)** neuron	conducts impulses
6	**(d)** sclerenchyma	fibres
7	**(d)** Protista	unicellular eukaryotes
8	**(c)** Arthropoda	insects, spiders, crabs
9	**(d)** cross-pollination	genetic variation
10	**(a)** Low temperature slows down microbial growth	fungi need warmth
11	**(b)** 100×10^{-9} m	1 nm = 10^{-9} m
12	**(d)** can travel through vacuum	unlike sound waves

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Unicellular: one cell, e.g. amoeba; multicellular: many cells, e.g. humans (1+1).	1+1
14	They are dividing rapidly and need dense cytoplasm, so vacuoles are absent (2).	2
15	Body structure, cell structure, mode of nutrition, habitat, reproduction, genetic similarity (any four, ½ each).	2
16	Showed that life comes only from pre-existing life, not from non-living matter (2).	2
17	1 kW × 1 h = 1 kWh = 3.6×10^6 J (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Place equal pieces in water and in 20% salt solution (1); weigh before and after (1); the piece in water gains weight and in salt solution loses weight (1).	1+1+1
19	Makes study easier; shows relationships; helps identify and name new organisms; supports conservation (1+1+1).	1+1+1
20	Fusion of gametes from two parents (1); random combinations of chromosomes (1); variation helps survival and evolution (1).	1+1+1
21	Warming melts glaciers (1); sea level rises, flooding coasts (hydrosphere) (1); affects ecosystems in the biosphere (1).	1+1+1
22	Cutting, grafting, layering, tissue culture (any three) (2); quick production of identical plants with desired traits (1).	2+1
	OR: Pollen tube reaches the ovule and the male gamete fuses with the egg (1); zygote becomes the embryo; ovule becomes the seed (1); ovary becomes the fruit (1).	1+1+1
23	Troposphere: weather; stratosphere: ozone; mesosphere: meteors burn; thermosphere/exosphere: thin (3).	3
	OR: It lies in the tropical and sub-tropical regions and gets abundant sunshine all year (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Conducts water vs food; dead vs living; direction upward vs both ways; tracheids/vessels vs sieve tubes (3). (ii) Transpiration pull and cohesion of water in the xylem (2).	3+2
	OR: (i) Apical, lateral, intercalary with sites and functions (2). (ii) Cutting root tips of an onion stops root elongation (2). (iii) Thin walls, dense cytoplasm, large nucleus, no vacuoles (1).	2+2+1
25	(i) Shape, tilt, latitude and surface type (2). (ii) Pressure differences make air flow from high to low (1). (iii) Day: sea breeze; night: land breeze (2).	2+1+2

	OR: (i) GHGs trap re-radiated heat (2). (ii) Rise in average temperature due to excess GHGs (1). (iii) Melting of glaciers, sea-level rise, extreme weather, monsoon changes (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** chromoplasts	yellow, orange, red
2	**(b)** osmosis	dilute to concentrated
3	**(c)** connective tissue	fluid matrix
4	**(b)** tissue	level of organisation
5	**(d)** dense cytoplasm and a prominent nucleus	rapid division
6	**(c)** biodiversity	diversity of life
7	**(a)** fishes	aquatic
8	**(d)** algae (thallophyta)	found in water
9	**(c)** insects	attract pollinators
10	**(a)** anther	male gametes
11	**(a)** carbon dioxide, methane and water vapour	trap heat
12	**(d)** condensation	cloud formation

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	It encloses the cell and controls the entry and exit of substances (2).	2
14	The process by which newly formed cells become specialised to perform specific functions (2).	2
15	They photosynthesise and released oxygen, making the Earth suitable for life (2).	2
16	Asexual: one parent, identical offspring; sexual: two parents, offspring with variations (1+1).	1+1
17	CO_2 from burning fossil fuels; methane from livestock or paddy fields (1+1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Nucleus: absent vs present with membrane; organelles: no membrane-bound organelles vs present; size: smaller vs larger (1+1+1).	1+1+1
19	Complex tissue of tracheids, vessels, fibres and parenchyma (1.5); tracheids and vessels are dead, tubular and thick-walled (1.5).	1½+1½
20	Wind (dandelion), water (coconut), animals (burs) (1+1+1).	1+1+1
21	Power = 5 kW (1); energy = 5 × 2 = 10 kWh (1) = 3.6×10^7 J (1).	1+1+1
22	The spherical shape gives different angles of rays (1.5); the tilt changes the angle and length of days during the year (1.5).	1½+1½
	OR: Venus has a thick atmosphere of CO_2 with a strong greenhouse effect (3).	3
23	Both have their own DNA and ribosomes (1.5); both can make some of their own proteins and divide on their own (1.5).	1½+1½
	OR: Cell wall present only in plant cells; plastids only in plants; large central vacuole in plants (1+1+1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Labelled diagram showing cell wall, membrane, cytoplasm, nucleus, vacuole, chloroplast, mitochondria (3). (ii) Cell wall and plastids (1). (iii) Chloroplast (1).	3+1+1
	OR: (i) Energy; transport network; sorting and packaging; digestion; protein synthesis (3). (ii) Mitochondria and plastids (1). (iii) Ribosomes (½). (iv) Lysosomes (½).	3+1+½+½
25	(i) Variety of life (1). (ii) Maintains ecosystems, food, medicines and stability (2). (iii) Decomposers recycle nutrients; plants provide food and oxygen (2).	1+2+2

	OR: (i) Thallophyta (Spirogyra), Bryophyta (moss), Pteridophyta (fern), Gymnosperms (pine), Angiosperms (mango) (3). (ii) Features of each (2).	3+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** mitosis	growth and repair
2	**(a)** ATP	adenosine triphosphate
3	**(a)** intercalary meristem	at the nodes/base
4	**(a)** 12	attached to the sternum
5	**(c)** Carolus Linnaeus	binomial nomenclature
6	**(d)** an alga and a fungus	bioindicators of air pollution
7	**(d)** with a capital first letter, in italics	e.g. Homo sapiens
8	**(d)** the egg is not fertilised	lining sheds
9	**(d)** 28 days	21–35 days
10	**(a)** tubers (eyes)	underground stem
11	**(b)** uneven heating of the Earth's surface	pressure differences
12	**(a)** 400×10^{-9} m	1 nm = 10^{-9} m

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	All living things are made of one or more cells; the cell is the basic unit of structure and function; new cells arise from pre-existing cells (2).	2
14	Loss of water vapour through stomata (1); creates transpiration pull and removes wastes (1).	1+1
15	Grouping of organisms on the basis of similarities and differences (1); to study the vast diversity systematically (1).	1+1
16	Light, single-celled reproductive units (1); to increase the chance of reaching suitable places (1).	1+1
17	The Sun's rays slant and spread over a larger area (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Plants cannot move and need rigid support to stay upright (1.5); animal cells need flexibility for movement and changing shape (1.5).	1½+1½
19	Collenchyma gives flexibility to the young stem (1.5); sclerenchyma fibres in coconut husk are rigid and lignified (1.5).	1½+1½
20	Seeds protect the embryo and aid dispersal (1.5); fruits enclose seeds and help dispersal by animals and wind (1.5).	1½+1½
21	Absorbs harmful UV; keeps the Earth warm by the greenhouse effect (1.5+1.5).	1½+1½
22	Receives, modifies, sorts and packages proteins and lipids into vesicles for transport and secretion (3).	3
	OR: Made of lipids and proteins (1); molecules can move sideways, flip and rotate (1); arranged like tiles in a mosaic (1).	1+1+1
23	Roots would starve (1); the phloem is the transporting tissue; the plant would die (2).	1+2
	OR: Single layer of protective cells (1); stomata in leaves for gas exchange and transpiration (1); root hairs for absorption (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) The outermost protective layer (1). (ii) Pores for gas exchange with guard cells controlling opening (2). (iii) Loss of water vapour; transpiration pull, cooling and removal of waste (2).	1+2+2
	OR: (i) Loosely packed living cells; corner-thickened living cells; dead lignified fibres and sclereids (3). (ii) Storage; flexibility; strength (2).	3+2
25	(i) Most pollen is lost (1). (ii) Sticky or spiny, large (1). (iii) To attract pollinators (1). (iv) Coral tree or hibiscus (2).	1+1+1+2

	OR: (i) A piece of stem with nodes is planted in moist soil (2). (ii) Joining a shoot to a rooted stock (1). (iii) Faster growth, identical plants, seedless varieties (2).	2+1+2

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